PART IIA, PAPER 3, SUPERVISION 0
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SOLUTIONS TO PROBLEMS

S0.1 Wondering how Part I Paper 3 was marked? This is definitely not it.
The Economics Faculty wishes to estimate the quantitative ability, θ, of a Part I student on the basis of a weighted combination, Z, of his Paper 3 Maths score, M, and Paper 3 Stats score, S:

$$Z = \lambda M + (1-\lambda) S. \tag{1}$$

Both M and S are random variables with mean θ; their variances are 4 and 9 respectively, and the correlation between them is ρ.

(a) Show that Z is an unbiased estimator of ability.
Answer: We want to show $\mathbb{E}[Z] = \theta$:

$$\mathbb{E}[Z] = \mathbb{E}[\lambda M + (1-\lambda) S] = \lambda\theta + (1-\lambda)\theta = \theta.$$

(b) Find the variance of Z.
Answer: Use the information from the question to compute

$$\begin{aligned}\mathrm{Var}(Z) &= \mathrm{Var}(\lambda M + (1-\lambda) S)\\ &= \lambda^2 \mathrm{Var}(M) + (1-\lambda)^2 \mathrm{Var}(S) + 2\lambda(1-\lambda)\mathrm{Cov}(M,S),\end{aligned}$$

where we know the variances of M and S, but not their covariance, so compute:

$$\rho = \mathrm{Corr}(M,S) = \frac{\mathrm{Cov}(M,S)}{\sqrt{\mathrm{Var}(M)\mathrm{Var}(S)}} \implies \mathrm{Cov}(M,S) = 6\rho.$$

Putting these results together, conclude that

$$\mathrm{Var}(Z) = 4\lambda^2 + 9(1-\lambda)^2 + 12\rho\lambda(1-\lambda) = \lambda^2(13-12\rho) + \lambda(12\rho - 18) + 9.$$

(c) Find the value of λ that minimizes the variance of the estimator when $\rho = 0$, and when $\rho = 0.5$. Check the second-order condition. Interpret these results.
Answer: Solve the following unconstrained minimization problem, where variance $\mathrm{Var}(Z|\lambda)$ is expressed explicitly as a function of the weight parameter:

$$\lambda^* = \arg\min_{\lambda} \mathrm{Var}(Z|\lambda) = \arg\min_{\lambda} \left[\lambda^2(13-12\rho) + \lambda(12\rho - 18) + 9\right].$$

Find the first order condition with respect to λ:

$$\frac{\partial}{\partial\lambda}\mathrm{Var}(Z|\lambda) = \lambda^*(26-24\rho) + (12\rho - 18) = 0 \implies \lambda^*(\rho) = \frac{9-6\rho}{13-12\rho}.$$

Check that this is indeed the minimum by computing the SOC:

$$\frac{\partial^2}{\partial\lambda^2}\mathrm{Var}(Z|\lambda) = 26 - 24\rho > 0, \quad \rho \in [-1,1].$$

$\mathrm{Var}(Z)$ is concave, hence $\lambda^*(\rho)$ is the global minimum. Substitute the particular values for ρ into $\lambda^*(\rho)$ to find that $\lambda^*(0) = \frac{9}{13}$ and $\lambda^*(0.5) = \frac{6}{7}$. The higher the correlation between the Maths and the Stats test scores, the higher the weight on the Maths score (with lower variance) should be in order to estimate quantitative ability.

S0.2 Bernoulli does Brexit: a European model for the British referendum. Consider the 2016 Brexit referendum where 48% of the population voted to remain.

(a) Explain why the voting decision of an individual i can be described by iid Bernoulli random variables $X_i \sim \text{iid}\,\text{Ber}(p)$, $i = 1, \ldots, n$. Write down the probability mass functions of X_i, derive its expectation and variance.

Answer: Start with the Bernoulli random variable $X_i \sim \text{Ber}(p)$, such that

$$X_i = \begin{cases} 1 & \text{if voted to remain} \\ 0 & \text{if voted to leave} \end{cases}$$

$X_i = 1$ with probability $p = 0.48$, for all i, which is also the population proportion of voters for remain. If the person voted leave, then $X_i = 0$ with probability $1-p = 0.52$. The pmf, the expectation and variance of X_i are given by:

$$\begin{aligned} \mathbb{P}(X_i = x) &= p^x (1-p)^{1-x} \\ \mathbb{E}[X_i] &= 1p + 0(1-p) = p \\ \text{Var}(X_i) &= \mathbb{E}\left[X_i^2\right] - (\mathbb{E}[X_i])^2 = 1^2 p - 0^2 (1-p) - p^2 = p(1-p) \end{aligned}$$

(b) What random variable would you use to model the total number of people who voted to remain? Call this random variable Y. Write down the probability mass function of Y, derive its expectation and variance. Explain what random variable you would use to model the proportion of remain voters. Derive its expectation and variance.

Answer: The number of people who voted to remain is the sum of X_i. We know that a sum of n iid Bernoulli random variables, $Y = \sum_{i=1}^n X_i$, has a Binomial distribution. The pmf of $Y \sim B(n, p)$ is given by

$$\mathbb{P}(Y = y) = \binom{n}{y} p^y (1-p)^{n-y}, \quad y = 0, 1, \ldots, n. \tag{2}$$

Use the expectation and variance of X_i's and their independence to compute:

$$\begin{aligned} \mathbb{E}[Y] &= \mathbb{E}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \mathbb{E}[X_i] = np, \\ \text{Var}(Y) &= \text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) + \sum_{i \neq j} \text{Cov}(X_i, X_j) = np(1-p), \end{aligned}$$

since independence of X_i's implies $\text{Cov}(X_i, X_j) = 0$ for all $i \neq j$. Similarly, the proportion of remain voters is the sample mean $\bar{X}_n$:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i, \qquad \mathbb{E}\left[\bar{X}_n\right] = p, \qquad \text{Var}\left(\bar{X}_n\right) = \frac{p(1-p)}{n}.$$

(c) When the sample size is large, the Binomial distribution can be approximated by the normal. This is a consequence of the central limit theorem (CLT). Explain.

Answer: The Central Limit Theorem says that if the mean and variance of a distribution exist and are finite, the sample mean of independent draws from the distribution

converges in distribution to a normal distribution. In other words, if X_i are iid with $\mathbb{E}[X_i] = \mu < \infty$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$, then the sample mean $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$ has the following property:

$$\bar{X}_n \overset{a}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right) \iff \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \overset{a}{\sim} N(0,1). \tag{3}$$

where $\overset{a}{\sim}$ means "approximately (asymptotically) distributed as". The approximation is close for samples of size greater than 30. We can apply the CLT to the sample mean from (b), $\bar{X}_n$, since it is a sum of n iid Bernoulli random variables. Note that in this case the mean is $\mu = p$ and the variance is $\sigma^2 = p(1-p)/n$, hence

$$Z = \frac{\bar{X}_n - p}{\sqrt{\frac{p(1-p)}{n}}} \overset{a}{\sim} N(0,1) \iff \mathbb{P}(Z \le z) \approx \Phi(z), \tag{4}$$

where $\Phi(\cdot)$ is the cdf of the standard normal random variable Z.

(d) Find the probability that a poll of (i) 200 people (ii) 1000 people, selected randomly, showed a majority of votes to leave the EU.

Answer: The population mean is $p = 0.48$. The sample proportion, $\hat{p} = \frac{1}{n}\sum_{i=1}^n X_i$, which is the same thing as the sample mean $\bar{X}_n$, has $\mathbb{E}[\hat{p}] = p$ and $\mathrm{Var}(\hat{p}) = p(1-p)/n$. Since $n > 30$ in both cases, according to the CLT:

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} \overset{a}{\sim} N(0,1) \iff \mathbb{P}(Z \le z) = \Phi(z).$$

We can then compute the required probabilities:

$$\mathbb{P}(\hat{p} < 0.5 \,|\, p = 0.48, n) = \Phi\left(\frac{0.5 - 0.48}{\sqrt{\frac{0.48 \times 0.52}{n}}}\right). \tag{5}$$

Use the standard normal tables to find that for $n = 200$, $\mathbb{P}(\hat{p} < 0.5) = 0.714$; and for $n = 1000$, $\mathbb{P}(\hat{p} < 0.5) = 0.897$.

(e) A poll of 300 voters from Cambridge South and 200 voters from Cambridge North showed that 56% and 48%, respectively, voted to remain. At a 5% significance level, test and calculate the p-value for the following hypotheses: (i) there is a difference in the support levels for remain between the two constituencies; (ii) remain had greater support in Cambridge South.

Answer: Denote the sample values given in the question by $n_S = 300$, $n_N = 200$, $\hat{p}_S = 0.56$, $\hat{p}_N = 0.48$, and the population parameters by p_S and p_N respectively. We want to test $H_0 : p_S = p_N$ against $H_1 : p_S \neq p_N$ or alternatively that $H_0 : d = p_S - p_N = 0$ and conduct inference on the random variable $\hat{d} = \hat{p}_S - \hat{p}_N$, which is the estimator of the difference between the sample proportions. Under H_0, $p_S = p_N = p$ and so $\mathbb{E}[\hat{d}] = 0$. We need to use the pooled sample mean $\hat{p}$ to estimate the standard error of $\hat{d}$. First, compute $\hat{p}$:

$$\hat{p} = \frac{0.56 \times 300 + 0.48 \times 200}{200 + 300} = 0.528.$$

It can be shown that

$$\operatorname{Var}\left(\hat{d}\right) = p\left(1-p\right)\left(\frac{1}{n_S}+\frac{1}{n_N}\right)$$

$$s_{\hat{d}} = \sqrt{\hat{p}\left(1-\hat{p}\right)\left(\frac{1}{n_S}+\frac{1}{n_N}\right)} = 0.0456$$

where $s_{\hat{d}}$ is the estimated standard error (estimate of the standard deviation of the sample mean). Due to the CLT we can conduct a t-test at $\alpha = 0.05$ using the following test statistic T and its asymptotic distribution under $H_0 : d = 0$

$$\hat{d} \overset{a}{\sim} N\left(0, s_{\hat{d}}^2\right) \iff T = \frac{\hat{d}-0}{s_{\hat{d}}} \overset{a}{\sim} N\left(0,1\right). \tag{6}$$

Use the standard normal critical value $z_{0.975} = 1.96$ to reject H_0 if $|t| > 1.96$, where t is the sample test statistic. Compute the sample test statistic:

$$t = \frac{\hat{d}}{s_{\hat{d}}} = \frac{0.56-0.48}{0.0456} = 1.756 < 1.96 \implies \text{do not reject } H_0 \text{ at } \alpha = 0.05.$$

The two-sided p-value is given by

$$p = \mathbb{P}\left(|T| > 1.756\right) = 2\Phi\left(-1.756\right) = 0.0793.$$

so that we would reject H_0 for any significance above this value.
For the one-sided test, use the same procedure as before, but with a different critical value, $z_{0.95} = 1.645 < 1.756 = t$, and we do reject H_0 at $\alpha = 0.05$. The one-sided p-value can be computed as

$$p = \mathbb{P}\left(T > 1.756\right) = 1 - \Phi\left(1.756\right) = 0.0396.$$

S0.3 When random variables unite they lose their independence.
Let X_j, $j = 1, 2, \ldots$, be a sequence of iid random variables with mean μ and variance σ^2. For $i = 1, 2, \ldots$ consider the random variable $Y_i = \sum_{j=1}^{i} X_j$.

(a) Find $\mathbb{E}\left[Y_i\right]$, $\operatorname{Var}\left(Y_i\right)$, $\operatorname{Cov}\left(Y_i, Y_k\right)$ for $i < k$ stating any properties being used.
Answer: Note that $Y_i = X_1 + \cdots + X_i$ and since X_j are iid, meaning $\operatorname{Cov}\left(X_k, X_l\right) = 0$ for $k \neq l$, then it is easy to see that $\mathbb{E}\left[Y_i\right] = i\mu$ and $\operatorname{Var}\left(Y_i\right) = i\sigma^2$. The covariance $\operatorname{Cov}\left(Y_i, Y_k\right)$ is a bit more tricky. First, note that Y_k is not independent of Y_i for $i < k$, because

$$Y_k = \sum_{j=1}^{k} X_j = \left(X_1 + \cdots + X_i\right) + \left(X_{i+1} + \cdots + X_k\right) = Y_i + \sum_{j=i+1}^{k} X_j.$$

Every Y_k depends on all previous Y_i for $i < k$, for instance $Y_k = Y_{k-1} + X_k$. We compute covariance as follows

$$\operatorname{Cov}\left(Y_k, Y_i\right) = \operatorname{Cov}\left(Y_i + \sum_{j=i+1}^{k} X_j, Y_i\right) = \operatorname{Cov}\left(Y_i, Y_i\right) + \operatorname{Cov}\left(\sum_{j=i+1}^{k} X_j, Y_i\right) = i\sigma^2$$

since the first term is $\mathrm{Cov}\,(Y_i, Y_i) = \mathrm{Var}\,(Y_i) = i\sigma^2$ and the second term is zero as X_i are iid and so $\mathrm{Cov}\,(X_j, X_i) = 0$ for all $i \neq j$, so that

$$\mathrm{Cov}\left(\sum_{j=i+1}^{k} X_j, Y_i\right) = \mathrm{Cov}\,(X_{i+1} + \cdots + X_k, X_1 + \cdots + X_i) = 0.$$

(b) State formally and provide intuition for the law of large numbers and the central limit theorem for iid random variables. Do these theorems apply to X_j or Y_i?

Law of large numbers (LLN). Let $X_1, \ldots, X_n$ be a sequence of iid random variables with expectation $\mathbb{E}\,[X_i] = \mu < \infty$ and variance $\mathrm{Var}\,(X_i) = \sigma^2 < \infty$. Then the sample mean, $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$, converges (in probability) to the expected value (population mean) μ as $n \to \infty$, i.e. $\bar{X}_n \to \mu$. This is illustrated in Figure 1. A sequence of iid random variables $X_i \sim N\,(0,1)$ is generated for $i = 1, \ldots, n$. The corresponding sample means, $\bar{X}_n$, are computed for each sample size between 1 and n. These sample means are plotted against the sample size together with a horizontal line that depicts the population mean $\mu = 0$. Note that as the sample size grows, the sample mean lies closer to the population mean.

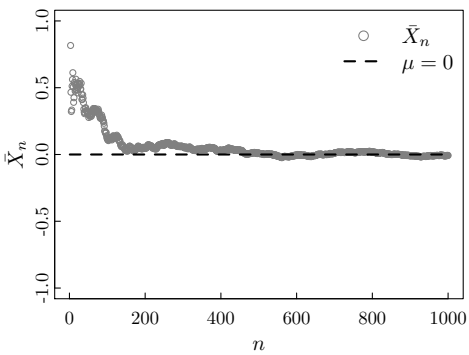

Figure 1: Law of large number (LLN) in action.

Central limit theorem (CLT). Let $X_1, \ldots, X_n$ be a sequence of iid random variables with expectation $\mathbb{E}\,[X_i] = \mu$ and variance $\mathrm{Var}\,(X_i) = \sigma^2 < \infty$, $i = 1, \ldots, n$. Then the distribution of the sample mean $\bar{X}_n$ converges to a normal distribution as $n \to \infty$. Four equivalent ways to write this formally:

$$\bar{X}_n \overset{a}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right) \quad \Longleftrightarrow \quad \sqrt{n}\left(\bar{X}_n - \mu\right) \overset{a}{\sim} N\left(0, \sigma^2\right)$$

$$\sqrt{n}\left(\frac{\bar{X}_n - \mu}{\sigma}\right) \overset{a}{\sim} N\,(0,1) \quad \Longleftrightarrow \quad \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \overset{a}{\sim} N\,(0,1)$$

That does not mean that $\bar{X}_n$ is exactly normal. Rather, this result holds only approximately and only in large samples, we sometimes say that $\bar{X}_n$ is asymptotically normal. This is illustrated in Figure 2, where we apply the CLT to a sequence of continuous, uniformly distributed random variables $X_i \sim \text{iid}\, U\,[-2, 2]$. For sample

sizes $n = 1, 2, 5, 10$, we generate one million samples of that corresponding size, compute $\bar{X}_n$ for each sample and plot the histogram of $\sqrt{n}\left(\bar{X}_n - \mu\right)/\sigma$ where $\mu = 0$ and $\sigma = 4/3$ are the corresponding mean and variance of the uniform distribution. The plot also shows the density function of the standard normal distribution (dashed line), which is the limiting distribution of $\sqrt{n}\left(\bar{X}_n - \mu\right)/\sigma$ according to the CLT. We can see that as n increases, the random variable $\sqrt{n}\left(\bar{X}_n - \mu\right)/\sigma$ gets much closer to $N(0, 1)$. Note that even for samples as small as $n = 10$ the match is pretty close.

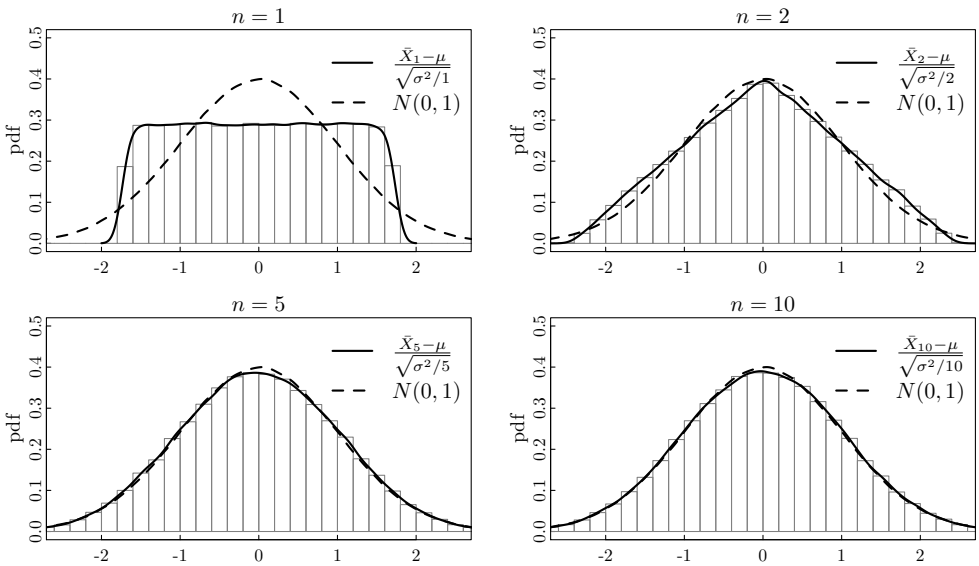

Figure 2: Central limit theorem (CLT) in action. X_i are iid $U[-2, 2]$, s.t. $\mu = \mathrm{E}[X_i] = 0, \sigma^2 = \mathrm{Var}(X_i) = 4/3$. Sample mean $\bar{X}_n = n^{-1}\sum_{i=1}^{n} X_i$, for sample sizes $n = 1, 2, 5, 10$.

LLN and CLT can be applied to the sample mean of $X_1, \ldots, X_n$ in the usual way since X_j are iid. However, as we discovered in (a), Y_i are not independent, so not iid, and hence we cannot apply the LLN and CLT to a random sample $Y_1, \ldots, Y_n$.

(c) Consider now a random variable Z_i defines as:

$$Z_i = \sum_{j=1}^{i} \omega_j X_j, \tag{7}$$

where ω_j is a sequence of real numbers (weights) satisfying two conditions:

$$\sum_{j=1}^{i} \omega_j = i, \qquad \sum_{j=1}^{i} \omega_j^2 > i \tag{8}$$

Which estimator would you prefer to estimate the population mean μ: $\bar{Y} = (Y_n/n)$ or $\bar{Z} = (Z_n/n)$? Note that $\bar{Z}$ is a general linear estimator of μ, while $\bar{Y}$ is a specific

case of $\bar{Z}$ when the weights for all observations are equal, $\omega_j = \frac{1}{n}$ for all j. [Hint: Compute expectation and variance of $\bar{Y}$ and $\bar{Z}$.]

Answer: Both estimators are linear, as they are linear combinations of the data observations. Intuitively, the difference between the two estimators is that $\bar{Y}$ is linear and assigns equal weights to every observations X_i, while $\bar{Z}$ is a more general linear estimator that may assign any weights to X_i, as long as the sum of these weights is equal to the number of the observations. Notice that $\bar{Y}$ is the usual sample mean taken from a sample of iid random variables $X_1, \ldots, X_n$. To compare $\bar{Y}$ and $\bar{Z}$, let's first check whether these estimators are unbiased:

$$\mathbb{E}\left[\bar{Y}\right] = \mathbb{E}\left[\frac{Y_n}{n}\right] = \frac{1}{n}\mathbb{E}\left[\sum_{j=1}^{n} X_j\right] = \frac{1}{n}\sum_{j=1}^{n}\mathbb{E}\left[X_j\right] = \frac{1}{n}n\mu = \mu,$$

$$\mathbb{E}\left[\bar{Z}\right] = \mathbb{E}\left[\frac{Z_n}{n}\right] = \frac{1}{n}\mathbb{E}\left[\sum_{j=1}^{n} \omega_j X_j\right] = \frac{1}{n}\sum_{j=1}^{n}\omega_j\mathbb{E}\left[X_j\right] = \frac{1}{n}n\mu = \mu.$$

Since both are unbiased, we would prefer the one that is more efficient, i.e. has lower variance. Hence, compute their variances, and recall that all cross-covariance terms $\mathrm{Cov}\left(X_i, X_j\right) = 0$, for $i \neq j$, since X_j are iid:

$$\mathrm{Var}\left(\bar{Y}\right) = \mathrm{Var}\left(\frac{Y_n}{n}\right) = \frac{1}{n^2}\mathrm{Var}\left(\sum_{j=1}^{n} X_j\right) = \frac{1}{n^2}\sum_{j=1}^{n}\mathrm{Var}\left(X_j\right) = \frac{\sigma^2}{n},$$

$$\mathrm{Var}\left(\bar{Z}\right) = \mathrm{Var}\left(\frac{Z_n}{n}\right) = \frac{1}{n^2}\mathrm{Var}\left(\sum_{j=1}^{n} \omega_j X_j\right) = \frac{1}{n^2}\sum_{j=1}^{n}\omega_j^2\mathrm{Var}\left(X_j\right) = \frac{\sigma^2}{n^2}\sum_{j=1}^{n}\omega_j^2,$$

since $\mathrm{Var}\left(X_j\right) = \sigma^2$. Now, recall that $\sum_{j=1}^{i}\omega_j^2 > i$, to conclude that

$$\mathrm{Var}\left(\bar{Z}\right) = \left(\frac{\sigma^2}{n^2}\right)\sum_{j=1}^{n}\omega_j^2 > \left(\frac{\sigma^2}{n^2}\right)n = \frac{\sigma^2}{n} = \mathrm{Var}\left(\bar{Y}\right)$$

Both estimators are linear, both are unbiased, but since $\bar{Y}$ has smaller variance, it is clearly a better estimator. In fact, $\bar{Y}$ is the best linear unbiased estimator, since no other linear unbiased estimator (in our case represented and generalized by $\bar{Z}$) has variance smaller than $\mathrm{Var}(\bar{Y})$. This result is saying pretty much the same thing as what the Gauss-Markov theorem says about the OLS estimators being best, i.e. having lowest variance in a class of linear unbiased estimators.

(d) Let $\sigma^2 = 1$. A random sample is drawn and the following statistic is obtained: $\bar{Y} = (Y_n/n) = (Y_{300}/300) = 2.79$. Test, at 5% significance, whether $\mu = 3$, against the alternative that $\mu \neq 3$, using three alternative methods: the t-statistic, the p-value and the confidence interval .

Answer: Since $\bar{Y}$ is the sample mean from a sample of iid random variables $X_1, \ldots, X_n$, we can safely apply the CLT to $\bar{Y}$:

$$\frac{\bar{Y} - \mu}{\sigma/\sqrt{n}} \overset{a}{\sim} N\left(0, 1\right).$$

Conduct a test of $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$, where the hypothesized value of the population mean is $\mu = 3$. Use the fact that under H_0 the following test statistic is approximately normal:

$$T = \frac{\bar{Y} - \mu_0}{\sigma/\sqrt{n}} \overset{a}{\sim} N(0,1).$$

Reject the null at $\alpha = 0.05$ if the absolute value of the sample test statistic t is greater than the critical value $z_{1-\frac{\alpha}{2}} = z_{0.975} = 1.96$. The observed sample test statistic is

$$t = \frac{2.79 - 3}{1/\sqrt{300}} = -3.637 \implies |t| > 1.96 \implies \text{reject } H_0 \text{ at } \alpha = 0.05.$$

The two-sided p-value is given by

$$p = \mathbb{P}(|T| > |t|) = 2\Phi(-3.637) = 0.0003 < \alpha,$$

and the null is rejected at 5%. As for the confidence interval, strictly speaking, we can only construct an asymptotic confidence interval for μ, since the distribution of X_j is not normal and hence we need to invoke the CLT in the above derivation of the distribution of T, even thought the variance of X_j is known. Note that since $n = 300$ is pretty large, we can be fairly confident that our approximation will be extremely close. Proceed in the usual way:

$$\mathrm{CI}_{0.95}(\mu) = \left[\bar{Y} - z_{0.975}\frac{\sigma}{\sqrt{n}}, \bar{Y} + z_{0.975}\frac{\sigma}{\sqrt{n}}\right] = [2.677, 2.903].$$

This says that there is a 95% chance that this interval contains the population mean, or that $\mathbb{P}(2.677 < \mu < 2.903) = 0.95$. Since the hypothesized mean $\mu_0 = 3$ lies outside $\mathrm{CI}_{0.95}(\mu)$, the null is rejected at 5%. The three testing procedures are equivalent and yield the same conclusion.

(e) Suppose now that for the same sample, the estimate of the population variance σ^2 is given by $S_{n-1}^2 = 1.11$. Test, at $\alpha = 0.05$, the hypothesis that $\sigma^2 = 1$.

Answer: We test $H_0 : \sigma^2 = \sigma_0^2$ against $H_1 : \sigma^2 > \sigma_0^2$, where the hypothesized value of the population variance is $\sigma_0^2 = 1$. Note that the alternative is one-sided, which is typical for testing variances. However, a two-sided hypothesis would have been perfectly fine too. Since the population mean is unknown and was estimated using the sample mean $\bar{Y}$, the unbiased estimator of the population variance is given by

$$S_{n-1}^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(Y_i - \bar{Y}\right)^2.$$

In this case, under the assumption that the null hypothesis is true, that is $\sigma^2 = \sigma_0^2$, the following random variable V, which equal to the estimator of the variance divided by population variance, corrected for the degrees of freedom $n-1$, has a chi-squared distribution with $n-1$ degrees of freedom:

$$V = \frac{(n-1)S_{n-1}^2}{\sigma_0^2} \sim \chi_{n-1}^2.$$

The test is one-sided and we reject H_0 at $\alpha = 0.05$ if $V > \chi^2_{0.95,299}$, which is the critical value of the χ^2_{299} distribution. Compute the sample test statistic:

$$v = \frac{299 \times 1.11}{1} = 331.89 < 340.32 = \chi^2_{0.95,299}$$

Hence, we cannot reject H_0 at $\alpha = 0.05$. Note that the p-value for this test is

$$p = \mathbb{P}(V > 331.89) = 1 - \mathbb{P}(V \leq 331.89) = 1 - 0.9075 = 0.0925,$$

where $\mathbb{P}(V \leq v)$ is the cdf of the random variable V, that is the cdf of the chi-squared distribution with 299 degrees of freedom. We can use statistical tables for χ^2_{299} to find that

$$\mathbb{P}(V \leq 331.89) = \chi^2_{299}(331.89) = 0.9075.$$

Based on $p = 0.0925$, we would reject the null at $\alpha = 0.1$.

S0.4 By the end of the year, you should be doing this in your sleep.
Consider a linear population regression, where x_i are fixed numbers and the residuals u_i are iid random variables with zero mean and constant variance σ^2, $u_i \sim \text{iid}\left(0, \sigma^2\right)$:

$$y_i = \beta_0 + \beta_1 x_i + u_i, \quad i = 1, \ldots, n. \tag{9}$$

(a) State the Gauss-Markov theorem with all of the assumptions.

Answer: The Gauss-Markov theorem states that the OLS estimators are Best Linear Unbiased Estimator (BLUE) under the following set of assumptions:

(A1) Linearity: the population model is linear in parameters, as in (9).

(A2) Fixed regressor: x_i are non-random and non-identical for all $i = 1, \ldots, n$.

(A3) Exogeneity: $\mathbb{E}[u_i|x_i] = 0$, for all $i = 1, \ldots, n$.

(A4) Homoscedasticity: $\text{Var}(u_i|x_i) = \sigma^2$, for $i = 1, \ldots, n$.

(A5) No serial correlation: $\text{E}[u_i u_j | x_i, x_j] = 0$ for $i \neq j$.

(b) Derive the OLS estimators of the parameters β_0 and β_1.

Answers: The residual is given by $u_i = y_i - \beta_0 - \beta_1 x_i$. We want to find the values of the coefficients that minimize the sum of squared residuals, SSR. We denote this SSR by $\phi(b_0, b_1)$, which is a function of the coefficients b_0 and b_1 that are allowed to vary freely. We call the values of the coefficients b_0, b_1 that minimize $\phi(b_0, b_1)$ the OLS estimators and denote them by $\hat{\beta}_0$ and $\hat{\beta}_1$, respectively. We have to solve the following problem:

$$\min_{b_0, b_1} \phi(b_0, b_1) = \min_{b_0, b_1} \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i)^2. \tag{10}$$

An illustration of what this problem looks like graphically is presented in Figure 3. It plots a hypothetical example where the population coefficients $(\beta_0, \beta_1) = (2, 2)$ and the free varying parameters in (10) are allowed to take values in the interval $[0, 4] \times [0, 4]$. You can see that $\phi(b_0, b_1)$ is a 3D quadratic function, where the xy-plane is defined by the pairs (b_0, b_1), and the z coordinate is the value of $\phi(b_0, b_1)$ corresponding to (b_0, b_1). The OLS looks for the pair $(b_0, b_1) = (\hat{\beta}_0, \hat{\beta}_1)$ that minimizes

the value of that function. Notice that the plot of the SSR function is increasing along the b_0-axis and the b_1-axis in all directions away from the OLS estimator that should be somewhere around the true value $(2, 2)$. This is saying that this problem has a unique solution.

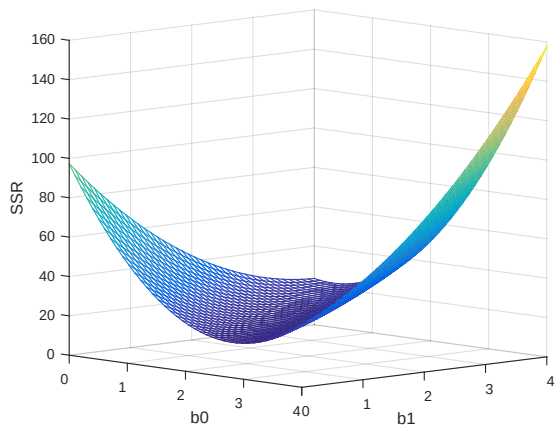

Figure 3: SSR as a function of the free-varying parameters, $\phi(b_0, b_1)$, from (10).

Algebraically, we can solve this problem by computing the first order conditions with respect to b_0 and b_1:

$$\frac{\partial \phi}{\partial b_0} = -2 \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i) = 0 \tag{11}$$

$$\frac{\partial \phi}{\partial b_1} = -2 \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i) x_i = 0 \tag{12}$$

The values of b_0 and b_1 that satisfy the first order conditions are precisely our OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$. Note that since the OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ satisfy both first order conditions in (11) and (12), so can replace these in the notation above. The following simplified expressions for (11) and (12) are called normal equations. Start with the first normal equation:

$$\sum_{i=1}^{n} \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i \right) = 0 \tag{13}$$

$$\sum_{i=1}^{n} y_i - n\hat{\beta}_0 - \hat{\beta}_1 \sum_{i=1}^{n} x_i = 0$$

Simple re-arranging of this expression yields

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}. \tag{14}$$

where $\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$ is the sample mean of x, and $\bar{y} = \frac{1}{n} \sum_{i=1}^{n} y_i$ is the sample mean of y. Now, substitute (14) into equation (12) and evaluate the second normal

equation:

$$\sum_{i=1}^{n}\left(y_i - \bar{y} - \hat{\beta}_1\bar{x} + \hat{\beta}_1 x_i\right) x_i = 0 \tag{15}$$

$$\sum_{i=1}^{n}\left(y_i - \bar{y}\right) x_i - \hat{\beta}_1 \sum_{i=1}^{n}\left(x_i - \bar{x}\right) x_i = 0.$$

After some re-arranging:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}\left(y_i - \bar{y}\right)\left(x_i - \bar{x}\right)}{\sum_{i=1}^{n}\left(x_i - \bar{x}\right)^2} = \frac{S_{xy}}{S_{xx}}, \tag{16}$$

where S_{xy} is the sample covariance of x and y, and S_{xx} the sample variance of x. An example of an OLS fit is presented in Figure 4.

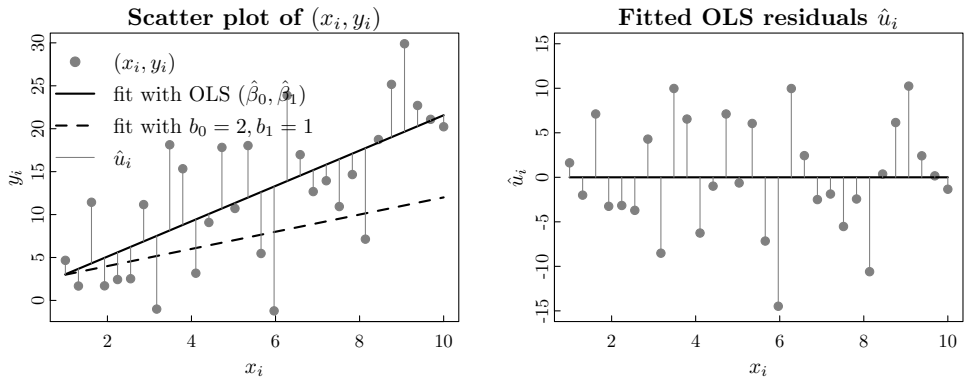

Figure 4: Illustration of OLS fit and residuals

The left panel presents a hypothetical scatter plot for (x_i, y_i) in gray dots, the solid line is the OLS line of fit, which is equal to $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$, and the corresponding OLS fitted residuals $\hat{u}_i = \hat{y}_i - y_i$, which are presented by the gray solid lines, and are equal to the distance from the data points to the fitted values $\hat{y}_i$ on the OLS line. What OLS essentially does is minimize the sum of squares of these gray distances. In addition to the OLS line of fit, another line of fit for some randomly chosen estimates values of b_0 and b_1 is presented as a dashed line. Note that these estimates (or any other possible pair of estimates) produces a line of fit that is strictly worse than the OLS estimates at minimizing the sum of squared residuals. The OLS fitted residuals are also plotted in the right panel against x_i. Note that $\sum_{i=1}^{n} \hat{u}_i = 0$ by construction of the OLS, and so the mean of $\hat{u}_i$ is zero.

(c) Show that the OLS estimator of β_1 can be expressed as

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}\left(x_i - \bar{x}\right) y_i}{\sum_{i=1}^{n}\left(x_i - \bar{x}\right)^2}. \tag{17}$$

Answer: Show that the numerator in (16) can be written as

$$\begin{aligned}\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y}) &= \sum_{i=1}^{n} y_i x_i - \sum_{i=1}^{n} y_i\bar{x} - \sum_{i=1}^{n}\bar{y}x_i + \sum_{i=1}^{n}\bar{y}\bar{x} \\ &= \sum_{i=1}^{n} y_i x_i - \sum_{i=1}^{n}\bar{y}x_i - n\bar{y}\bar{x} + n\bar{y}\bar{x} \\ &= \sum_{i=1}^{n}(x_i-\bar{x})\,y_i.\end{aligned}$$

Similarly, we can show that the denominator can be expressed as

$$\sum_{i=1}^{n}(x_i-\bar{x})^2 = \sum_{i=1}^{n}(x_i-\bar{x})\,x_i.$$

Intuitively, this means that the sample covariance can be computed by de-meaning only one of the variables. The two expressions combined yield one more alternative for writing $\hat{\beta}_1$:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i-\bar{x})\,y_i}{\sum_{i=1}^{n}(x_i-\bar{x})\,x_i}. \tag{18}$$

(d) Use your answer in (c) to show that $\hat{\beta}_1$ can be expressed as

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^{n}(x_i-\bar{x})\,u_i}{\sum_{i=1}^{n}(x_i-\bar{x})^2}. \tag{19}$$

Answer: Substitute $y_i = \beta_0 + \beta_1 x_i + u_i$ into (17):

$$\begin{aligned}\hat{\beta}_1 &= \frac{\sum_{i=1}^{n}(x_i-\bar{x})\,y_i}{\sum_{i=1}^{n}(x_i-\bar{x})^2} = \frac{\sum_{i=1}^{n}(x_i-\bar{x})(\beta_0+\beta_1 x_i+u_i)}{\sum_{i=1}^{n}(x_i-\bar{x})^2} \\ &= \beta_0\frac{\sum_{i=1}^{n}(x_i-\bar{x})}{\sum_{i=1}^{n}(x_i-\bar{x})^2} + \beta_1\frac{\sum_{i=1}^{n}(x_i-\bar{x})(x_i-\bar{x})}{\sum_{i=1}^{n}(x_i-\bar{x})^2} + \frac{\sum_{i=1}^{n}(x_i-\bar{x})\,u_i}{\sum_{i=1}^{n}(x_i-\bar{x})^2} \\ &= \beta_1 + \frac{\sum_{i=1}^{n}(x_i-\bar{x})\,u_i}{\sum_{i=1}^{n}(x_i-\bar{x})^2}.\end{aligned}$$

The first term in the second line is zero because

$$\sum_{i=1}^{n}(x_i-\bar{x}) = \sum_{i=1}^{n}x_i - \sum_{i=1}^{n}\bar{x} = n\bar{x} - n\bar{x} = 0.$$

This expression is very useful for studying the properties of the OLS estimators. It says that the OLS estimator is equal to the population parameter that it estimates plus an additional term that be expressed in terms of the sample covariance of the explanatory variable and the residual (S_{xu}) in the numerator, and the variance of the explanatory variable (S_{xx}) in the denominator:

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^{n}(x_i-\bar{x})\,u_i}{\sum_{i=1}^{n}(x_i-\bar{x})^2} = \frac{\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})\,u_i}{\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2} = \beta_1 + \frac{S_{xu}}{S_{xx}}.$$

Since the sample variance S_{xx} is strictly positive, the estimator will be equal to the population coefficient, $\hat{\beta}_1 = \beta_1$, only if the sample covariance of the explanatory variable and the residual is zero. This may be true in expectation, provided that we assume that the residual is independent from x, which is precisely assumption (A2) of exogeneity, or zero conditional mean of u (a lot more on this later). However, when working with real data in empirical applications, S_{xu} is never exactly zero, so that $\hat{\beta}_1$ will almost surely not be exactly equal to β_1.

(e) Use equation (19) to derive an expression for $\text{Var}(\hat{\beta}_1)$. Explain what the Gauss Markov theorem says about $\text{Var}(\hat{\beta}_1)$.

Answer: First, note that since β is a constant, $\text{Var}(\hat{\beta}) = \text{Var}(\hat{\beta} - \beta)$. With that in mind, take variance on both sides of expression (19):

$$\text{Var}\left(\hat{\beta}_1\right) = \text{Var}\left(\frac{\sum_{i=1}^n (x_i - \bar{x})\, u_i}{\sum_{i=1}^n (x_i - \bar{x})^2}\right) = \left(\frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2}\right)^2 \text{Var}\left(\sum_{i=1}^n (x_i - \bar{x})\, u_i\right)$$

where for simplicity of notation we denote the term in the denominator by

$$SST_x = \sum_{i=1}^n (x_i - \bar{x})^2\,, \tag{20}$$

which is the Total Sum of Squares of x. This is similar to the sample variance of x, S_{xx}, but not corrected for the number of observations. SST_x then measures the total variation in the explanatory variable, rather than the average variation, which is measured by S_{xx}. Use this notation in the above expression to simplify:

$$\begin{aligned}
\text{Var}\left(\hat{\beta}_1\right) &= \frac{1}{SST_x^2}\text{Var}\left(\sum_{i=1}^n (x_i - \bar{x})\, u_i\right) \\
&= \frac{1}{SST_x^2}\text{Var}\left(\sum_{i=1}^n (x_i - \bar{x})\, u_i\right) \\
&= \frac{1}{SST_x^2}\text{Var}\Big((x_1 - \bar{x})\, u_1 + (x_2 - \bar{x})\, u_2 + \cdots + (x_n - \bar{x})\, u_n\Big)\Big) \\
&= \frac{1}{SST_x^2}\left(\sum_{i=1}^n (x_i - \bar{x})^2\, \text{Var}\,(u_i) + 2\sum_{i<j}^n (x_i - \bar{x})\,(x_j - \bar{x})\, \text{Cov}\,(u_i, u_j)\right)
\end{aligned}$$

We can simplify this expression by using the Gauss-Markov assumptions of homoscedasticity: $\text{Var}\,(u_i) = \sigma^2$ for all i, as well as the assumption of no serial-correlation: $\text{Cov}\,(u_i, u_j) = 0$ for all $i \neq j$,

$$\text{Var}\left(\hat{\beta}_1\right) = \frac{1}{SST_x^2}\left(\sigma^2 \sum_{i=1}^n (x_i - \bar{x})^2\right) = \frac{\sigma^2}{SST_x} = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}. \tag{21}$$

The Gauss Markov theorem says that $\hat{\beta}_1$ is efficient, that is it has the lowest variance out of all linear unbiased estimators of β. For example, if $\tilde{\beta}_1$ is some other linear unbiased estimator of β, that is $\mathbb{E}[\tilde{\beta}_1] = \beta_1$, then the $\text{Var}(\tilde{\beta}_1) \geq \text{Var}(\hat{\beta}_1)$. Look back at the Q3(c), does that look familiar?

S0.5 Confident balls dye hard.
An urn contains 5 white and 10 black balls. A fair die is rolled and a number of balls corresponding to the number shown on the die is randomly chosen from the urn.

(a) Find the probability that all the balls selected are white.
Answer: Let A denote the event that all balls are white and B_j the event that the die landed showing number $j = \{1, 2, 3, 4, 5, 6\}$. Use the Law of Total Probability and the fact that $\mathbb{P}(B_j) = \frac{1}{6}$ for all j, to show that

$$\begin{aligned}\mathbb{P}(A) &= \sum_{j=1}^{6} \mathbb{P}(A \cap B_j) = \sum_{j=1}^{6} \mathbb{P}(A|B_j)\,\mathbb{P}(B_j) = \mathbb{P}(B_j) \sum_{j=1}^{6} \mathbb{P}(A|B_j) \\ &= \frac{1}{6}\left(\frac{5}{15} + \frac{5}{15}\frac{4}{14} + \frac{5}{15}\frac{4}{14}\frac{3}{13} + \frac{5}{15}\frac{4}{14}\frac{3}{13}\frac{2}{12} + \frac{5}{15}\frac{4}{14}\frac{3}{13}\frac{2}{12}\frac{1}{11}\right) \\ &= 0.076.\end{aligned}$$

(b) Find the conditional probability that the die landed showing a 3, if all the balls selected are white.
Answer: Using the Bayes' rule for conditional probability

$$\mathbb{P}(B_3|A) = \frac{\mathbb{P}(A|B_3)\,\mathbb{P}(B_3)}{\mathbb{P}(A)} = \frac{\left(\frac{5}{15}\frac{4}{14}\frac{3}{13}\right)\frac{1}{6}}{0.076} = 0.048.$$

(c) Suppose the proportion of while balls in the urn is not known and needs to be estimated. To do so, draw a random sample with replacement. For $i = 1, \ldots, n$, let $X_i = 1$ if the ball is white, and 0 if black. Consider the following estimators:

$$\hat{p}_n = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad \tilde{p}_n = X_n$$

Find $\mathbb{E}[\cdot]$ and $\mathrm{Var}(\cdot)$ of $\hat{p}_n$ and $\tilde{p}_n$. Which estimator is preferable and why?
Answer: Observe that X_i, $i = 1, \ldots, n$, are iid Bernoulli random variables, with $\mathbb{E}[X_i] = p$ and $\mathrm{Var}(X_i) = p(1-p)$. We can use these expressions to find:

$$\begin{aligned}\mathbb{E}[\hat{p}_n] &= \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i] = p. \\ \mathbb{E}[\tilde{p}_n] &= \mathbb{E}[X_n] = p. \\ \mathrm{Var}(\hat{p}_n) &= \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = \frac{p(1-p)}{n} \\ \mathrm{Var}(\tilde{p}_n) &= \mathrm{Var}(X_n) = p(1-p).\end{aligned}$$

In order to compute $\mathrm{Var}(\hat{p}_n)$ we used independence of X_i's, i.e. $\mathrm{Cov}(X_i, X_j) = 0$ for all $i \neq j$. Hence, in the full expansion of the third line above we obtain that

$$\mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = \mathrm{Var}(X_1 + \cdots + X_n) = \sum_{i=1}^{n}\mathrm{Var}(X_i) + 2\sum_{i<j}\mathrm{Cov}(X_i, X_j)$$

Clearly, both $\hat{p}_n$ and $\tilde{p}_n$ are unbiased. Since $\text{Var}(\hat{p}_n) \leq \text{Var}(\tilde{p}_n)$, for $n \geq 1$, and with strict inequality for $n > 1$, $\hat{p}_n$ is more efficient. Intuitively, this is because $\hat{p}_n$ uses more observations. In fact, $\hat{p}_n$ is the BLUE of p.

(d) Construct a confidence interval for the true proportion p using $\hat{p}_n$.
Answer: According to the Central Limit theorem, as $n \to \infty$

$$\hat{p}_n \sim N\left(p, \frac{p(1-p)}{n}\right) \iff Z = \frac{\hat{p}_n - p}{\sqrt{\frac{p(1-p)}{n}}} \overset{a}{\sim} N(0,1). \tag{22}$$

However, since we don't know the population parameter p, we don't know the variance of X_i either. We can estimate $\text{Var}(X_i)$ using $\hat{p}_n(1-\hat{p}_n)$, so that the estimator of the variance of $\hat{p}_n$ is $\hat{p}_n(1-\hat{p}_n)/n$. The asymptotic distribution is

$$T = \frac{\hat{p}_n - p}{\sqrt{\frac{\hat{p}_n(1-\hat{p}_n)}{n}}} \overset{a}{\sim} N(0,1). \tag{23}$$

Pick a significance level α. Then the $(1-\alpha)100\%$ asymptotic confidence interval for p, $\text{CI}_{1-\alpha}(p)$, where the population proportion lies with probability $1-\alpha$, or in other words satisfies $\mathbb{P}\left(-z_{1-\frac{\alpha}{2}} \leq p \leq z_{1-\frac{\alpha}{2}}\right) = 1-\alpha$. Use the critical values of the standard normal distribution, $\Phi^{-1}\left(1-\frac{\alpha}{2}\right) = z_{1-\frac{\alpha}{2}}$, to compute

$$\text{CI}_{1-\alpha}(p) = \left[\hat{p} - z_{1-\frac{\alpha}{2}}\sqrt{\frac{\hat{p}_n(1-\hat{p}_n)}{n}}, \hat{p} + z_{1-\frac{\alpha}{2}}\sqrt{\frac{\hat{p}_n(1-\hat{p}_n)}{n}}\right]. \tag{24}$$

(e) Knowing that $\hat{p}_{100} = 0.25$, test at 5% the hypothesis that the proportion of white balls in the urn is 0.3 against the alternative that (i) it is not; (ii) it is greater.
Answer: We want to conduct inference on the population proportion p, so construct the following hypotheses $H_0 : p = p_0$ against $H_1 : p \neq p_0$, where the hypothesized value of the population proportion $p_0 = 0.3$. Under the null hypothesis, we know that according to the CLT, the sample proportion $\hat{p}$ computed from the sample of size $n = 100$ is normally distributed around the population proportion (under the null $p = p_0$):

$$\hat{p}_n \sim N\left(p_0, \frac{p_0(1-p_0)}{n}\right) \iff T = \frac{\hat{p}_n - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} \sim N(0,1). \tag{25}$$

This test statistic follows a standard normal distribution. Note that under the null, the standard error of $\hat{p}_n$ is determined by the population proportion, assumed known under the null hypothesis, that is $p = p_0$. The rejection rule for the two-sided test with, say, $\alpha = 0.05$, is $|t| > z_{0.975} = 1.96$, where t is the sample test statistic. Compute the sample test-statistic with $\hat{p}_{100} = 0.25$:

$$t = \frac{0.25 - 0.3}{\sqrt{\frac{0.3(1-0.3)}{100}}} = -1.09 \implies |t| < 1.96, \text{ do not reject } H_0 \text{ at } 5\%.$$

For (ii), use the same idea as in (i), but now the alternative is one-sided, so test $H_0 : p = 0.3$ against $H_1 : p > 0.3$. The critical value for the test with $\alpha = 0.05$ is now

$z_{0.95} = 1.645$ and since $|t| < 1.645$, we do not reject the null at 5% significance. The is sufficient evidence that the true proportion of white balls in the urn is not greater than 0.3.

S0.6 Reconciling population differences with large independent samples.
Consider a sample of iid random variables, $X_i \sim \text{iid}\left(\mu, \sigma^2\right)$, of size n.

(a) Prove that the sample mean is an unbiased estimator of the population mean, and the variance of the sample mean is σ^2/n.

Answer: Denote the sample mean by $\bar{X}_n = n^{-1}\sum_{i=1}^n X_i$ and compute:

$$\mathbb{E}\left[\bar{X}_n\right] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^n X_i\right] = \frac{1}{n}\mathbb{E}\left[X_1 + \cdots + X_n\right] = \frac{1}{n}\sum_{i=1}^n \mathbb{E}\left[X_i\right] = \frac{1}{n} n\mathrm{E}\left[X_1\right] = \mu,$$

$$\mathrm{Var}\left(\bar{X}_n\right) = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^n X_i\right) = \frac{1}{n^2}\left(\sum_{i=1}^n \mathrm{Var}\left(X_i\right) + 2\sum_{i<j}\mathrm{Cov}\left(X_i, X_j\right)\right) = \frac{\sigma^2}{n},$$

The covariances disappear as X_i are iid and hence $\mathrm{Cov}\left(X_i, X_j\right) = 0$, for all $i \neq j$.

(b) Suppose $\bar{X}_1$ and $\bar{X}_2$ are sample means for two large independent samples of sizes n_1 and n_2 from populations with means μ_1 and μ_2 and variances σ_1^2 and σ_2^2 respectively. Determine the distribution of $\hat{d} = \bar{X}_1 - \bar{X}_2$ and explicitly state any assumptions that you make.

Answer: Since the samples are large ($n \geq 30$), according to the Central Limit theorem for iid random variables

$$\bar{X}_1 = \frac{1}{n_1}\sum_{i=1}^{n_1} X_i^{(1)} \overset{a}{\sim} N\left(\mu_1, \frac{\sigma_1^2}{n_1}\right), \qquad \bar{X}_2 = \frac{1}{n_2}\sum_{j=1}^{n_2} X_j^{(2)} \overset{a}{\sim} N\left(\mu_2, \frac{\sigma_2^2}{n_2}\right),$$

where $X_i^{(1)}$, $i = 1, \ldots, n_1$, and $X_j^{(2)}$, $j = 1, \ldots, n_2$, are the random variables from the first and the second sample, respectively. The difference of the means is normal since the samples are independent, hence

$$\hat{d} = \bar{X}_1 - \bar{X}_2 \overset{a}{\sim} N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right).$$

The variance is computed as

$$\begin{aligned}
\mathrm{Var}\left(\bar{X}_1 - \bar{X}_2\right) &= \mathrm{Var}\left(\frac{1}{n_1}\sum_{i=1}^{n_1} X_i^{(1)} + \frac{1}{n_2}\sum_{j=1}^{n_2} X_j^{(2)}\right) \\
&= \mathrm{Var}\left(\frac{1}{n_1}\sum_{i=1}^{n_1} X_i^{(1)}\right) + \mathrm{Var}\left(\frac{1}{n_2}\sum_{j=1}^{n_2} X_j^{(2)}\right) \\
&+ \mathrm{Cov}\left(\frac{1}{n_1}\sum_{i=1}^{n_1} X_i^{(1)}, \frac{1}{n_2}\sum_{j=1}^{n_2} X_j^{(2)}\right)
\end{aligned}$$

The crucial assumption here is that the samples are independent, hence the covariance terms $\text{Cov}(X_i^{(1)}, X_j^{(2)}) = 0$ for all i and j.

$$\begin{aligned}
\text{Var}\left(\bar{X}_1 - \bar{X}_2\right) &= \text{Var}\left(\frac{1}{n_1}\sum_{i=1}^{n_1} X_i^{(1)}\right) + \text{Var}\left(\frac{1}{n_2}\sum_{j=1}^{n_2} X_j^{(2)}\right) \\
&= \left(\frac{1}{n_1}\right)^2 \sum_{i=1}^{n_1} \text{Var}\left(X_i^{(1)}\right) + \left(\frac{1}{n_2}\right)^2 \sum_{j=1}^{n_2} \text{Var}\left(X_j^{(2)}\right) \\
&= \left(\frac{1}{n_1}\right)^2 \sum_{i=1}^{n_1} \sigma_1^2 + \left(\frac{1}{n_2}\right)^2 \sum_{j=1}^{n_2} \sigma_2^2 \\
&= \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}
\end{aligned}$$

(c) The summary statistics for average outpatient waiting times (in weeks) for 152 Primary Care Trusts in England, in March 2007, and March 2008 are given below:

	# PCTs	**Mean**	**St.Dev.**
London PCTs 2008	31	2.82	0.528
Other PCTs 2008	121	2.57	0.406
All PCTs 2008	152	2.62	0.443
All PCTs 207	152	3.77	0.477
Change 2008-2007	152	-1.44	0.534

Use a 1% significance level to test the hypothesis that there was no difference between PCT waiting times for trusts in London and elsewhere in March 2008. Find and interpret the p-value associated with your test.

Answer: Denote the population mean of London PCT's 2008 by μ_L and the population mean for other PCTs as μ_O. We are interested to test $H_0 : \mu_L = \mu_O$ against $H_1 : \mu_L \neq \mu_O$ at $\alpha = 0.01$. Note that the two samples can be assumed to be independent, since they represent different PCTs. Then, use sample information from the table above and the distribution we found in (b) to compute:

$$\hat{d} = \bar{X}_L - \bar{X}_O \overset{a}{\sim} N\left(\mu_L - \mu_O, \frac{\sigma_L^2}{n_L} + \frac{\sigma_O^2}{n_O}\right) \overset{H_0}{=} N\left(0, \frac{0.528^2}{31} + \frac{0.406^2}{121}\right)$$

since under H_0, $\mu_L - \mu_O = 0$. Hence, $\hat{d} \overset{H_0}{\sim} N\left(0, 0.1018^2\right)$ and use the following test statistic to conduct inference on H_0:

$$T = \frac{\hat{d}}{s_{\hat{d}}} \sim N(0, 1) \tag{26}$$

where the standard error is given by

$$s_{\hat{d}} = \sqrt{\widehat{\text{Var}}\left(\hat{d}\right)} = 0.1018$$

Reject H_0 at 1% if the absolute value of the following test statistic is greater than the critical value $z_{0.995} = 2.57$:

$$t = \frac{2.82 - 2.57}{0.1018} = 2.456 < 2.57 \implies \text{do not reject } H_0 \text{ at } 1\%.$$

The p-value for this two-sided test is given by

$$\mathbb{P}\left(|T| > |t|\right) = 2\left(1 - \Phi\left(2.456\right)\right) = 0.014$$

so the test is not rejected for any significance level below $\alpha = 0.014$.

(d) Suppose that you want to test the hypothesis that PCT waiting times decreased on average between March 2007 and March 2008. Explain why, in this case, it might not be appropriate to estimate the variance of the difference between means, $\hat{d}$, from the standard deviations given in lines 3 and 4.

Answer: This is because the samples are not independent, as the 2008 sample consists of the same PCTs as the 2007 sample, so the values for individual PCTs are likely to be highly (positively) correlated, meaning that the computation of $\text{Var}(\hat{d})$ in (b) would not only involve the variances of individual observations from each sample, but also covariance between the observations, so the sum of the individual variances is likely to underestimate the true variance.

(e) Line 5 in the table shows the summary statistics for the change in waiting times for each PCT between March 2007 and March 2008. Use this to test at a 1% significance level the hypothesis that waiting times fell over this period.

Answer: Test $H_0\colon d = 0$ against $H_1\colon d < 0$ where $d = \mu_{2008} - \mu_{2007}$. Use the following distribution of the estimator and the t-test statistic under the null:

$$\hat{d} \overset{H_0}{\sim} N\left(0, \frac{0.534^2}{152}\right) = N\left(0, 0.0433^2\right) \iff T = \frac{\hat{d}}{s_{\hat{d}}} \overset{H_0}{\sim} N\left(0, 1\right).$$

Reject H_0 at 1% if the sample statistics is smaller than $z_{0.001} = -2.33$:

$$t = \frac{-1.44}{0.0433} = -33.26 < -2.33 \implies \text{reject } H_0 \text{ at } 1\%.$$

S0.7 "The chicken or the egg" not so much a dilemma for the linear regression. The following table gives average annual growth rates for employment e_i and productivity p_i in the manufacturing industry for the period from 1953-54 to 1963-64 for a sample of $n = 12$ countries. The data for this question are also available in the `STATA` file `kaldor.dta`.

(a) Compute correlation between the growth rates e_i and p_i.

Answer: The sample correlation is given by

$$r_{e,p} = \frac{S_{ep}}{\sqrt{S_{ee}}\sqrt{S_{pp}}} = \frac{\sum_{i=1}^{n}\left(e_i - \bar{e}\right)\left(p_i - \bar{p}\right)}{\sqrt{\sum_{i=1}^{n}\left(e_i - \bar{e}\right)^2}\sqrt{\sum_{i=1}^{n}\left(p_i - \bar{p}\right)^2}}, \tag{27}$$

where S_{ep} is the sample covariance between e_i and p_i, S_{ee} is the sample variance of e_i, and S_{pp} is the sample variance of p_i:

$$S_{ep} = \widehat{\mathrm{Cov}}\left(e_i, p_i\right) = \frac{1}{n-1}\sum_{i=1}^{n}\left(e_i - \bar{e}\right)\left(p_i - \bar{p}\right) = 1.561, \tag{28}$$

$$S_{ee} = \widehat{\mathrm{Var}}\left(e_i\right) = \frac{1}{n-1}\sum_{i=1}^{n}\left(e_i - \bar{e}\right)^2 = 2.386, \tag{29}$$

$$S_{pp} = \widehat{\mathrm{Var}}\left(p_i\right) = \frac{1}{n-1}\sum_{i=1}^{n}\left(p_i - \bar{p}\right)^2 = 2.382. \tag{30}$$

Here $\bar{e} = 2.2$ and $\bar{p} = 3.9$ are the sample means of employment growth and productivity growth, respectively. Using data from Table 1 above, correlation is $r_{e,p} = 0.6547$.

Table 1: Employment growth (e_i) and productivity growth (p_i)

Country	e_i	p_i	$(e_i - \bar{e})^2$	$(p_i - \bar{p})^2$	$(e_i - \bar{e})(p_i - \bar{p})$
Austria	2.0	4.2	0.058	0.090	-0.073
Belgium	1.5	3.9	0.550	0.000	0.000
Canada	2.3	1.3	0.003	6.760	-0.152
Denmark	2.5	3.2	0.067	0.490	-0.181
France	1.9	3.8	0.117	0.010	0.034
Italy	4.4	4.2	4.658	0.090	0.648
Japan	5.8	7.8	12.662	15.210	13.878
Netherlands	1.9	4.1	0.117	0.040	-0.068
Norway	0.5	4.4	3.033	0.250	-0.871
West Germany	2.7	4.5	0.210	0.360	0.275
UK	0.6	2.8	2.695	1.210	1.806
US	0.8	2.6	2.078	1.690	1.874
Sum	26.9	46.8	26.249	26.200	17.170
Unbiased mean	2.2	3.9	2.386	2.382	1.561

(b) Compute the slope coefficients for the regressions of e_i on p_i, and p_i on e_i. Confirm that the product of the two slope coefficients is equal to the square of the correlation coefficient you found in (a).

Answer: We are estimating two regressions:

$$e_i = \beta_0 + \beta_1 p_i + u_i, \tag{31}$$

$$p_i = \gamma_0 + \gamma_1 e_i + v_i, \tag{32}$$

using OLS, where the corresponding estimators are given by

$$\hat{\beta}_1 = \frac{S_{ep}}{S_{pp}} = \frac{\frac{1}{n-1}\sum_{i=1}^{n}\left(p_i - \bar{p}\right)\left(e_i - \bar{e}\right)}{\frac{1}{n-1}\sum_{i=1}^{n}\left(p_i - \bar{p}\right)^2}, \tag{33}$$

$$\hat{\gamma}_1 = \frac{S_{ep}}{S_{ee}} = \frac{\frac{1}{n-1}\sum_{i=1}^{n}\left(p_i - \bar{p}\right)\left(e_i - \bar{e}\right)}{\frac{1}{n-1}\sum_{i=1}^{n}\left(e_i - \bar{e}\right)^2}. \tag{34}$$

The estimates of the slope coefficients are $\hat{\beta}_1 = 0.6553$ and $\hat{\gamma}_1 = 0.6541$. The scatter plots and the OLS fitted lines are presented in Figure 5. Notice that the two regression lines look almost identical, which confirms that the slopes are almost exactly the same. Checking the relationship between the slope estimates and the correlation coefficient

$$r_{e,p}^2 = \hat{\beta}_1 \hat{\gamma}_1 = 0.4286.$$

Note also that the intercept coefficients are given by

$$\hat{\beta}_0 = \bar{e} - \hat{\beta}_1 \bar{p} = \bar{e} - \left(\frac{S_{ep}}{S_{pp}}\right) \bar{p}, \tag{35}$$

$$\hat{\gamma}_0 = \bar{p} - \hat{\gamma}_1 \bar{e} = \bar{p} - \left(\frac{S_{ep}}{S_{ee}}\right) \bar{e}. \tag{36}$$

(c) Show, using the formulae for correlation and the regression slope coefficients, that the equality from (b) is true for a general bivariate relationship between any two variables x_i and y_i.

Answer: Using the formulae for the OLS estimator from (33) and (34), as well as the expression for the correlation coefficient from (27) compute

$$\hat{\beta}_1 \hat{\gamma}_1 = \left(\frac{S_{ep}}{S_{pp}}\right)\left(\frac{S_{ep}}{S_{ee}}\right) = \frac{S_{ep}^2}{S_{ee} S_{pp}} = r_{e,p}^2. \tag{37}$$

Notice that we can then write the OLS estimators as

$$\hat{\beta}_1 = \frac{S_{ep}}{S_{pp}} = \left(\frac{S_{ep}}{\left(\sqrt{S_{pp}}\right)^2}\right)\left(\frac{\sqrt{S_{ee}}}{\sqrt{S_{ee}}}\right) = \left(\frac{S_{ep}}{\sqrt{S_{pp}\sqrt{S_{ee}}}}\right)\sqrt{\frac{S_{ee}}{S_{pp}}} = r_{e,p}\sqrt{\frac{S_{ee}}{S_{pp}}}, \tag{38}$$

$$\hat{\gamma}_1 = \frac{S_{ep}}{S_{ee}} = \left(\frac{S_{ep}}{\left(\sqrt{S_{ee}}\right)^2}\right)\left(\frac{\sqrt{S_{pp}}}{\sqrt{S_{pp}}}\right) = \left(\frac{S_{ep}}{\sqrt{S_{pp}\sqrt{S_{ee}}}}\right)\sqrt{\frac{S_{pp}}{S_{ee}}} = r_{e,p}\sqrt{\frac{S_{pp}}{S_{ee}}}. \tag{39}$$

(d) Compute the t-ratios for the two slope coefficients and confirm that they are identical. Using the relevant formulae, explain why this is the case.

Answer: First, test the hypothesis $H_0 \colon \beta_1 = 0$ against $H_1 \colon \beta_1 \neq 0$. Since the sample size is small, we have to assume that the error terms from regression (31) are normal, $u_i \sim \text{iid}\, N\left(0, \sigma^2\right)$. However, this assumption is only required to conduct hypothesis testing, it will not affect the general result that the t-ratios from the two regressions are identical. We know that under the null hypothesis, the distribution of the OLS estimator is given by

$$T_{\hat{\beta}_1} = \frac{\hat{\beta}_1}{s_{\hat{\beta}_1}} \sim t_{n-2}, \tag{40}$$

where $s_{\hat{\beta}_1}$ is the standard error of the OLS estimator given by

$$s_{\hat{\beta}_1}^2 = \frac{\hat{\sigma}_u^2}{SST_p} = \frac{\hat{\sigma}_u^2}{\sum_{i=1}^n \left(p_i - \bar{p}\right)^2} = \frac{\hat{\sigma}_u^2}{(n-1)\, S_{pp}}. \tag{41}$$

Here SST_p is the total sum of squares or the total variability in p_i. Express the regression standard error or the root mean squared error (RMSE) in terms of the correlation coefficients:

$$
\begin{aligned}
\hat{\sigma}_u^2 &= \frac{1}{n-2}\sum_{i=1}^{n}\left(e_i - \hat{\beta}_0 - \hat{\beta}_1 p_i\right)^2 \\
&= \frac{1}{n-2}\sum_{i=1}^{n}\left(e_i - \left(\bar{e} - \left(\frac{S_{pe}}{S_{pp}}\right)\bar{p}\right) - \left(\frac{S_{pe}}{S_{pp}}\right)p_i\right)^2 \\
&= \frac{1}{n-2}\sum_{i=1}^{n}\left((e_i - \bar{e}) - \left(\frac{S_{pe}}{S_{pp}}\right)(p_i - \bar{p})\right)^2 \\
&= \frac{1}{n-2}\left[\sum_{i=1}^{n}(e_i - \bar{e})^2 + \sum_{i=1}^{n}\left(\left(\frac{S_{pe}}{S_{pp}}\right)(p_i - \bar{p})\right)^2 - 2\sum_{i=1}^{n}\left(\frac{S_{pe}}{S_{pp}}\right)(e_i - \bar{e})(p_i - \bar{p})\right] \\
&= \frac{1}{n-2}\left[\sum_{i=1}^{n}(e_i - \bar{e})^2 + \left(\frac{S_{pe}}{S_{pp}}\right)^2\sum_{i=1}^{n}(p_i - \bar{p})^2 - 2\left(\frac{S_{pe}}{S_{pp}}\right)\sum_{i=1}^{n}(e_i - \bar{e})(p_i - \bar{p})\right] \\
&= \frac{1}{n-2}\left[(n-1)S_{ee} + \left(\frac{S_{pe}}{S_{pp}}\right)^2(n-1)S_{pp} - 2\left(\frac{S_{pe}}{S_{pp}}\right)(n-1)S_{ep}\right] \\
&= \left(\frac{n-1}{n-2}\right)\left[S_{ee} - \frac{S_{ep}^2}{S_{pp}}\right] \\
&= \left(\frac{n-1}{n-2}\right)\left[S_{ee} - \frac{S_{ep}^2}{S_{pp}}\frac{S_{ee}}{S_{ee}}\right] \\
&= \left(\frac{n-1}{n-2}\right)\left[S_{ee} - S_{ee}\left(\frac{S_{ep}^2}{S_{pp}S_{ee}}\right)\right] \\
&= \left(\frac{n-1}{n-2}\right)\left[S_{ee} - S_{ee}r_{e,p}^2\right] \\
&= \left(\frac{n-1}{n-2}\right)\left(1 - r_{e,p}^2\right)S_{ee}
\end{aligned}
$$

That implies that the standard error of $\hat{\beta}_1$ is given by

$$
s_{\hat{\beta}_1}^2 = \left(\frac{n-1}{n-2}\right)\left(1 - r_{e,p}^2\right)S_{ee}\left[\frac{1}{(n-1)S_{pp}}\right] = \left(\frac{1 - r_{e,p}^2}{n-2}\right)\left(\frac{S_{ee}}{S_{pp}}\right). \tag{42}
$$

Using these results and the expression for $\hat{\beta}_1$ from (38) and $r_{e,p}$ from (27), we can derive

$$
T_{\hat{\beta}_1} = \frac{\hat{\beta}_1}{s_{\hat{\beta}_1}} = \left[r_{e,p}\sqrt{\frac{S_{ee}}{S_{pp}}}\right]\left[\sqrt{\frac{n-2}{1 - r_{e,p}^2}}\sqrt{\frac{S_{pp}}{S_{ee}}}\right] = r_{e,p}\sqrt{\frac{n-2}{1 - r_{e,p}^2}} = 2.739. \tag{43}
$$

Notice that the t-statistic only depends on the correlation coefficient, and hence it must be the case that $T_{\hat{\gamma}_1} = T_{\hat{\beta}_1}$. Computation using the data confirm that.

(e) What economic interpretation would you attach to your results?

Answer: The estimate of β suggests that an increase of one unit in the productivity growth rate increases the employment growth rate by 0.65534. The estimate for γ suggests that an increase of one unit in the employment growth rate implies an increase of 0.65412 in the productivity growth rate. We cannot say that these are causal relationships.

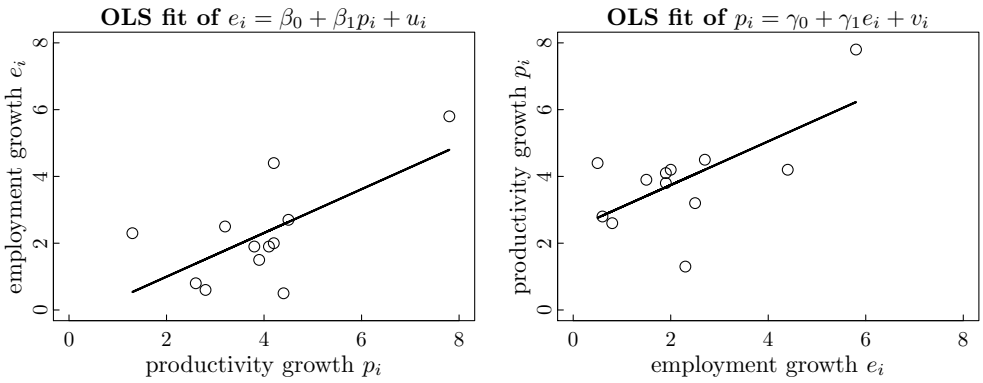

Figure 5: Scatter plots and OLS regression lines of fit